XRPL Validator — Daily Engineering Report · 2026-03-22

Period: 2026-03-22 00:00 to 24:00 US Eastern (2026-03-22 04:00 to 2026-03-23 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: git history only. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 9 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat(viewer): auto peer discovery from TMEndpoints** (`506923e`, xrpl-validator) When a connected peer sends TMEndpoints, we automatically connect to new peers we haven't seen before. Filters out local/IPv6 addrs. Max 15 concurrent peer connections. Peers at hop count <= 1 only.
- **feat(viewer): live peer count with connected/discovered display** (`861112a`, xrpl-validator) /api/peers endpoint returns connected count, known peers, and max. Header shows PEERS 2/5 (connected/discovered) updating every 5s. Peer panel shows all discovered peers sorted alphabetically.
- **fix: sled storage for sync, peer identity verification, consensus wiring** (`3a37944`, xrpl-validator) Three incomplete items from assessment, now fixed: 1. Sync writes to sled DB alongside JSONL - sled::Batch for atomic page writes - Binary key/value storage (not hex text) - DB at [local path]
- **fix(viewer): neural network consensus viz with tab-switch fix** (`24703dd`, xrpl-validator) Restore neural network visualization as the consensus page. When switching tabs and returning, accumulated pulses are now flushed (detects >0.5s frame gap) to prevent lag burst. SSE events skip queuing visual effects while tab is hidden.
- **feat(ledger): transaction application engine with testnet verification** (`c64f330`, xrpl-validator) Complete transaction engine implementing the XRPL ledger close pipeline: - Keylet system: deterministic 256-bit state keys (3 mainnet-verified) - Sandbox: transactional overlay with snapshot/rollback for tec safety - Transactor trait: preflight → preclaim → apply_common → do_apply - Payment: XRP direct payments with account creation, reserve checks - OfferCreate/Cancel: DEX order placement with crossing framework - …
- **feat(viewer): engine dashboard, tx-type visuals, regression tests** (`6818946`, xrpl-validator) Metrics page: - XRP fee burn counter with sparkline (total + per-round) - Transaction type breakdown bars (color-coded by type) - Engine coverage card (supported vs unsupported percentage)

- **feat(ledger): complete tx type coverage + engine dashboard + audited** (`ebaa41e`, xrpl-validator) Transaction types (33 total, up from 6): - Added: Escrow (Create/Finish/Cancel), Check (Create/Cash/Cancel), PayChannel (Create/Claim/Fund), NFToken (Mint/Burn/CreateOffer/ AcceptOffer/CancelOffer), AMM (Create/Deposit/Withdraw/Vote/Bid/ Delete), SetRegularKey, SignerListSet, DepositPreauth, Clawback, Credential (Create/Delete/Accept)
- **feat: live engine, peer network page, dashboard redesign, identity** (`343ae01`, xrpl-validator) Live Engine: - Opens sled DB (18.7M+ entries) and applies transactions to real mainnet state in real-time - Decodes binary XRPL objects, deducts fees, writes back to sled - Tracks applied/failed/total stats exposed via /api/engine - XRPL_NO_ENGINE env var disables engine while sync runs

Docs and operations

- **docs: add LICENSE (MIT), README with security warnings, .gitignore** (`90697c1`, xrpl-validator)

Independence program

No independence record for UTC day 2026-03-22; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
03-22 04:07	—	xrpl-validator	`506923e`	feat(viewer): auto peer discovery from TMEndpoints
03-22 04:11	—	xrpl-validator	`861112a`	feat(viewer): live peer count with connected/discovered display
03-22 04:41	—	xrpl-validator	`3a37944`	fix: sled storage for sync, peer identity verification, consensus wiring
03-22 04:59	—	xrpl-validator	`90697c1`	docs: add LICENSE (MIT), README with security warnings, .gitignore
03-22 23:38	—	xrpl-validator	`24703dd`	fix(viewer): neural network consensus viz with tab-switch fix

Committed	Merged into main	Repo	Commit	Subject
03-22 23:38	—	xrpl-validator	`c64f330`	feat(ledger): transaction application engine with testnet verification
03-23 00:02	—	xrpl-validator	`6818946`	feat(viewer): engine dashboard, tx-type visuals, regression tests
03-23 00:49	—	xrpl-validator	`ebaa41e`	feat(ledger): complete tx type coverage + engine dashboard + audited
03-23 02:34	—	xrpl-validator	`343ae01`	feat: live engine, peer network page, dashboard redesign, identity

Times in UTC. A commit appears on the day it was committed to main or merged into it.